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TABLE OF CONTENTS

1 RELATED DOCUMENTS **4**

2 DEFINITIONS AND ABBREVIATIONS **5**

3 ACTIVITIES TO BE PERFORMED **6**

3.1 Summary **7**

3.2 Limited Assay Qualification **9**

4 RESULTS FROM QUALIFICATION RUNS IN DOG TISSUE **9**

5 ASSAY PROCEDURE **21**

5.1 Preparation of Stock Solution **21**

5.1.1 Instruments and Materials 21

5.1.2 Reagents 21

5.1.3 Method 22

5.2 Preparation of the Assay Protocol **22**

5.3 Preparation of Working Solutions **23**

5.3.1 Instruments and Materials 23

5.3.2 Reagents 23

5.3.3 Method 23

5.4 Tissue Preparation **23**

5.4.1 Instruments and Materials 23

5.4.2 Reagents 24

5.4.3 Tissue Preparation 24

5.4.4 When the lysates have thawed to be processed 24

5.4.4.1 Sample Dilution 25

5.5 Hybridization **25**

5.5.1 Instruments and Materials 25

5.5.2 Reagents 25

5.5.3 Method 26

5.6 HPLC Analysis **26**

5.6.1 Material and Reagents 26

5.6.2 Instruments 27

5.6.3 Method 30

5.7 Evaluation of HPLC Chromatograms **31**

6 RECORDS **32**

7 CHANGE HISTORY 32

TABLE OF TABLES

Table 1 Parameters of final method 8
Table 2 Calibration Curve Accuracy in dog brain tissue 12
Table 3 Confirmation of LLOQ in dog tissue 13
Table 4 Selectivity in dog tissue 14
Table 5 Carry-over in dog brain tissue 15
Table 6 Accuracy and Precision in dog brain tissue 16
Table 7 Accuracy and Precision in dog liver tissue 17
Table 8 Accuracy and Precision in dog kidney tissue 18
Table 9 Freeze and thaw and short term stability in dog brain 19
Table 10 Dilution Integrity in dog brain tissue 20
Table 11 HPLC systems 27
Table 12 Gradient table 31

TABLE OF FIGURES

Figure 1 Structure and sequence of the analyte 2439exNA 6
Figure 2 HPLC-profile of 2439exNA in dog brain tissue at the LLOQ (20.0 ng/g) 7
Figure 3 HPLC-profile of 2439exNA in dog brain tissue at the ULOQ (10,000.0 ng/g) 7

1 RELATED DOCUMENTS

This instruction describes the execution of a PNA-HPLC-Assay based on the following documents:

- KU-RGL-SOP-002 (formerly: REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-002 (formerly: REG-102-S02) – Method Validation Data Report
  - KU-RGL-ESS-003 (formerly: REG-102-S03) – Method Validation Protocol PNA Assay - Solid Matrices
- KU-RGL-SOP-009 (formerly: REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-012 (formerly: REG-109-S02) – PNA Assay Instruction and Protocol - Solid Matrices
  - KU-RGL-ESS-014 (formerly: REG-109-S04) – Preparation of HPLC Buffers & Stock Solutions
  - KU-RGL-INS-005 (formerly: REG-109-I01) – PNA-HPLC-Assay SST
- Development protocol in the project folder AN401.0-24 under the following link:

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The above mentioned documents describe the assay execution under nonGLP-relevant conditions. The parameters and results described in this document are generated from a nonGLP method development, which was conducted following the above mentioned SOPs.

2 DEFINITIONS AND ABBREVIATIONS

ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
CoA	Certificate of Analysis
CV	Coefficient of Variation
DNA	Deoxyribonucleic Acid
ed	Edition
EDTA	Ethylenediaminetetraacetic Acid
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
HPLC	High Performance Liquid Chromatography
ICH	International Conference on Harmonization
IFU	Instructions For Use
LC/MS	Liquid Chromatography / Mass Spectrometry
LLOQ	Lower Limit of Quantitation
min	Minute
n.a.; N/A	Not applicable
PBS	Phosphate Buffered Saline
PI	Principal Investigator
PNA	Peptide Nucleic Acid
QC	Quality Control
rcf	Relative centrifugal force
RNA	Ribonucleic Acid
RT	Room temperature
SDS	Sodium Dodecyl Sulfate
siRNA	Small Interfering Ribonucleid Acid
SOP	Standard Operating Procedure
SD	Standard Deviation
SST	System Suitability Test(ing)
TCEP	Tris(2-carboxyethyl) phosphine hydrochloride solution
TCLS	Tissue & Cell Lysis Solution
ULOQ	Upper Limit of Quantitation

3 ACTIVITIES TO BE PERFORMED

This method development report describes an anion exchange (AEX)-HPLC method with fluorescence detection that allows the specific detection of the analyte 2439exNA from dog tissue. 2439exNA is a is a dimer composed of two identical siRNAs, which are linked at their sense strands with a tetraethylene glycol linker, with the sequence depicted in Figure 1:

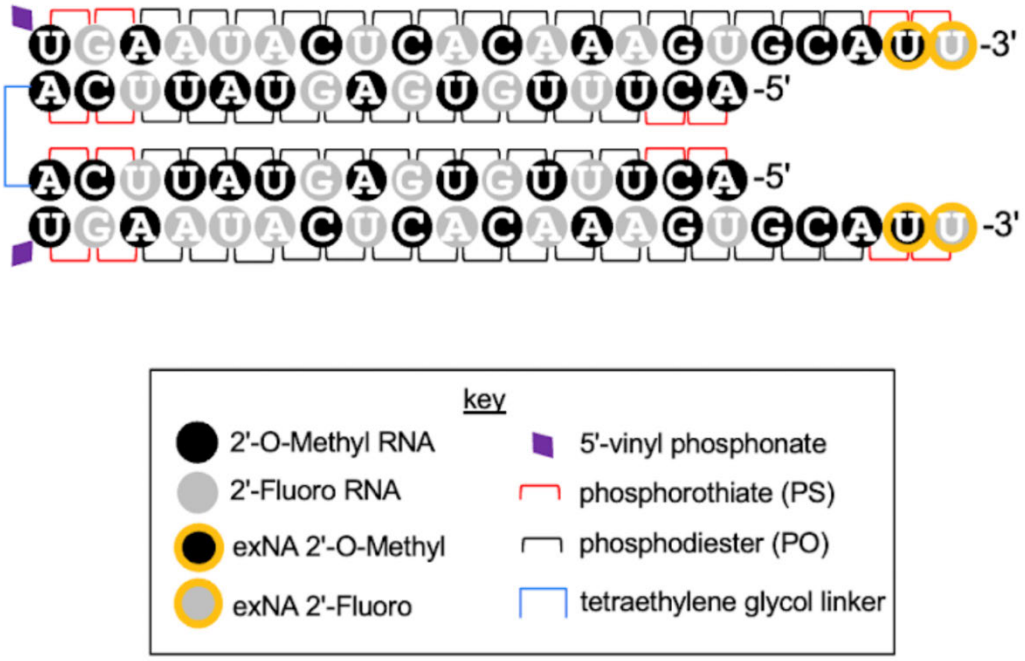

Figure 1 Structure and sequence of the analyte 2439exNA

The assay is based on the specific hybridization of the antisense strand of 2439exNA to the complementary PNA-probe. The PNA-probe is N-terminally conjugated with an Atto425-fluorescent dye. The utilized PNA-BroadInst-01 is complementary to the antisense strand of 2439exNA and has the following sequence:

N-term Atto425-OO-cdc ttt gtg dgt d C-term

This method was developed without having any metabolites of 2439exNA.

The method was developed for dog brain, liver and kidney tissue using dog brain as surrogate matrix.

Figure 2 and Figure 3 show the HPLC-profile of 2439exNA in dog brain tissue.

3.1 Summary

2. [redacted]

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4 Results from Qualification runs in dog tissue

In this section, the raw data for the determination of calibration curve accuracy (Table 2), confirmation of LLOQ (Table 3), selectivity (Table 4), carry-over (Table 5), accuracy and precision (Table 6), freeze and thaw and short term stability (Table 9) and dilution integrity (Table 10) from 2439exNA in dog tissue are shown. The results were generated in two qualifying runs. Calibration standard and QC samples were prepared by spiking 2439exNA into dog tissue. Samples were prepared according to the assay procedure described in Section 5.

During the qualification run, the calibration range was between 20.0 ng/g and 10,000.0 ng/g. All calibration standards were within ±15 % (±20 % at the LLOQ) of the nominal concentration.

Selectivity and Sensitivity

The LLOQ was confirmed at 20.0 ng/g in dog tissue. No signal was detected in corresponding blank tissue samples and the 2439exNA peaks at the LLOQ of 20.0 ng/g were at least 5 times the background signal in the analysis run. At the LLOQ, the mean back-calculated concentration was 102.6 % of nominal with a CV of 0.9 %.

The selectivity of the method was demonstrated by comparison of unspiked samples of dog tissue with the same tissue samples spiked with 2439exNA at a concentration of the LLOQ. The signal for

2439exNA in the spiked samples gave an at least 5-fold higher response (peak area) compared to the blank samples.

Carry Over

Carry over tissue samples were run after the 10,000.0 ng/g calibration standards. There were no distinct analyte signal detectable in the carry-over samples after the 10,000.0 ng/g standards.

Accuracy and Precision

QC samples were prepared at 4 concentration levels which are covering the calibration curve range:

- at the LLOQ (QC-LLOQ): 20.0 ng/g
- at 3 times above the LLOQ (QC-Low): 60.0 ng/g
- at around 50 % of the calibration curve range (QC-Mid): 4,500.0 ng/g
- at least at 75 % of the upper calibration curve range (QC-High): 10,000.0 ng/g

The QC samples for the determination of the accuracy and precision of the method met the acceptance criteria with following results:

Dog brain:

- 100 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.

AN401.0-24_01_AB08

- The mean accuracies at the 4 concentration levels were 101.8 % (QC-LLOQ), 99.7 % (QC-Low), 100.3 % (QC-Mid) and 99.8 % (QC-High).
- The mean precisions at the 4 concentration levels were 4.0 % (QC-LLOQ), 1.5 % (QC-Low), 0.1 % (QC-Mid) and 1.2 % (QC-High).

AN401.0-24_01_AB09

- The mean accuracies at the 3 concentration levels were 92.9 % (QC-Low), 97.7 % (QC-Mid) and 97.7 % (QC-High).
- The mean precisions at the 3 concentration levels were 0.1 % (QC-Low), 0.1 % (QC-Mid) and 0.7 % (QC-High).

Dog liver

- 100 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.
- The mean accuracies at the 4 concentration levels were 100.7 % (QC-LLOQ), 101.1 % (QC-Low), 100.0 % (QC-Mid) and 99.2 % (QC-High).
- The mean precisions at the 4 concentration levels were 5.8 % (QC-LLOQ), 1.7 % (QC-Low), 0.1 % (QC-Mid) and 0.2 % (QC-High).

Dog kidney

- 100 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.

- The mean accuracies at the 4 concentration levels were 102.8 % (QC-LLOQ), 101.1 % (QC-Low), 100.8 % (QC-Mid) and 99.2 % (QC-High).
- The mean precisions at the 4 concentration levels were 5.7 % (QC-LLOQ), 0.6 % (QC-Low), 0.2 % (QC-Mid) and 0.3 % (QC-High).

Freeze-Thaw and Short-Term-Stability

The freeze and thaw (one cycle) and short term stability of 2439exNA in dog tissue was successfully tested for up to 4 hours. Aliquots of the 2439exNA at the 2 QC concentration levels QC-Low and QC-High were stored at room temperature for 4 hours and analyzed in triplicate.

After 4 hours at room temperature the mean back-calculated concentrations were at 102.7 % (QC-Low) and 101.6 % (QC-High) of the nominal value.

Dilution Integrity

The precision and accuracy of dog tissue sample dilution was evaluated using a QC standard prepared at concentration levels 500-fold (5,000,000.0 ng/g) and above the ULOQ of 10,000.0 ng/g. This QC sample was analyzed and quantified after 1:1,000 dilution in 3 replicates. Samples were diluted with buffer solution (10 % ACN / 0.01 % Tween / 1x PBS / 1mM TCEP or 10% ACN / 1x PBS / 1mM TCEP).

- Buffer solution 10 % ACN / 0.01 % Tween / 1x PBS / 1mM TCEP: the mean accuracy for the 5,000,000.0 ng/g dilution standard (1:1,000) was 106.3 % and the CV was 1.9 %.
- Buffer solution 10 % ACN / 1x PBS / 1mM TCEP: the mean accuracy for the 5,000,000.0 ng/g dilution standard (1:1,000) was 96.1 % and the CV was 1.4 %.

Table 2 Calibration Curve Accuracy in dog brain tissue

2439exNA		Cal-Level-2		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7	
		20.0 ng/g		100.0 ng/g		500.0 ng/g		1,000.0 ng/g		5,000.0 ng/g		10,000.0 ng/g	
Run ID	Sample No.	Calc. 2439exNA Conc. ng/g	Acc. criterion +/-20 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal
AN401.0-24_01_AB08	5-10	20.4	102.0%	98.0	98.0%	485.8	97.2%	983.2	98.3%	4,914.8	98.3%	9,856.2	98.6%
AN401.0-24_01_AB08	91-96	20.4	102.0%	99.2	99.2%	501.6	100.3%	1,031.9	103.2%	5,074.3	101.5%	10,154.3	101.5%
AN401.0-24_01_AB09	4-9	20.8	104.0%	97.3	97.3%	493.5	98.7%	977.7	97.8%	4,956.5	99.1%	10,002.4	100.0%
AN401.0-24_01_AB09	29-34	20.5	102.5%	99.6	99.6%	500.0	100.0%	995.5	99.6%	5,044.3	100.9%	10,031.8	100.3%
Mean		20.53	102.6%	98.53	98.5%	495.23	99.0%	997.08	99.7%	4,997.48	99.9%	10,011.18	100.1%
SD (+/-)		0.19		1.06		7.19		24.38		74.41		122.48	
CV (%)		0.9%		1.1%		1.5%		2.4%		1.5%		1.2%	
N		4		4		4		4		4		4	

Table 3 Confirmation of LLOQ in dog tissue

2439exNA		Cal-Level-1			Cal-Level-2		
		0.0 ng/g			20.0 ng/g		
Run ID	Sample No.	Peak Area [mV*min]	Calc. 2439exNA Conc. ng/g	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of LLOQ
AN401.0-24_01_AB08	4, 5	0.0000	BQL	0.0%	0.0379	20.4	102.0%
AN401.0-24_01_AB08	90, 91	0.0000	BQL	0.0%	0.0381	20.4	102.0%
AN401.0-24_01_AB09	3, 4	0.0000	BQL	0.0%	0.0418	20.8	104.0%
AN401.0-24_01_AB09	28, 29	0.0000	BQL	0.0%	0.0412	20.5	102.5%
Mean		0.00000	n.a.	0.0%	0.03975	20.53	102.6%
SD (+/-)		0.00000	n.a.		0.00204	0.19	
CV (%)		n.a.	n.a.		5.1%	0.9%	
N		4	4		4	4	

Table 4 Selectivity in dog tissue

2439exNA					Blank		QC-LLOQ	
					0.0 ng/g		20.0 ng/g	
Run ID	Date	Sex	Lot #	Sample No.	Peak Area [mV*min]	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439exNA Conc. ng/g
AN401.0-24_01_AB08 Brain	16-Oct-2024	n.a.	Broad	13, 25	0.0000	0.0%	0.0366	19.8
	16-Oct-2024	n.a.	Broad	14, 26	0.0000	0.0%	0.0372	20.0
	16-Oct-2024	n.a.	Broad	15, 27	0.0000	0.0%	0.0401	21.3
AN401.0-24_01_AB08 Liver	16-Oct-2024	n.a.	Broad	17, 29	0.0000	0.0%	0.0352	19.1
	16-Oct-2024	n.a.	Broad	18, 30	0.0000	0.0%	0.0368	19.90
	16-Oct-2024	n.a.	Broad	19, 31	0.0000	0.0%	0.0402	21.4
AN401.0-24_01_AB08 Kidney	16-Oct-2024	n.a.	Broad	21, 33	0.0000	0.0%	0.0366	19.7
	16-Oct-2024	n.a.	Broad	22, 34	0.0000	0.0%	0.0414	21.9
	16-Oct-2024	n.a.	Broad	23, 35	0.0000	0.0%	0.0374	20.1

Table 5 Carry-over in dog brain tissue

2439exNA		Cal-Level-2			Cal-Level-7			Carry Over		
		20.0 ng/g			10,000.0 ng/g			0.0 ng/g		
Run ID	Sample No.	Peak Area [mv*min]	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Peak Area [mv*min]	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Peak Area [mv*min]	Calc. 2439exNA Conc. ng/g	Acc. Criterion ≤ 20 % of LLOQ
AN401.0-24_01_AB08	5, 10, 11	0.0379	20.4	102.0%	22.3591	9,856.2	98.6%	0.0000	BQL	0.0%
AN401.0-24_01_AB08	91, 96, 97	0.0381	20.4	102.0%	23.0357	10,154.3	101.5%	0.0000	BQL	0.0%
AN401.0-24_01_AB09	4, 9, 10	0.0418	20.8	104.0%	22.7741	10,002.4	100.0%	0.0000	BQL	0.0%
AN401.0-24_01_AB09	29, 34, 35	0.0412	20.5	102.5%	22.8411	10,031.8	100.3%	0.0000	BQL	0.0%

Table 6 Accuracy and Precision in dog brain tissue

2439exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	Sample No.	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal
AN401.0-24_01_AB08	25, 37, 49, 61	19.8	99.0%	59.1	98.5%	4,508.3	100.2%	9,088.3	101.0%
	26, 38, 50, 62	20.0	100.0%	59.5	99.2%	4,516.0	100.4%	8,974.8	99.7%
	27, 39, 51, 63	21.3	106.5%	60.8	101.3%	4,513.9	100.3%	8,877.1	98.6%
Mean		20.37	101.8%	59.80	99.7%	4,512.73	100.3%	8,980.07	99.8%
SD (+/-)		0.81		0.89		3.98		105.70	
CV (%)		4.0%		1.5%		0.1%		1.2%	
N		3		3		3		3	
AN401.0-24_01_AB09	12-14	n.a.	n.a.	55.8	93.0%	4,395.8	97.7%	8,747.9	97.2%
	24-26	n.a.	n.a.	55.7	92.8%	4,399.2	97.8%	8,835.7	98.2%
Mean		n.a.	n.a.	55.75	92.9%	4,397.50	97.7%	8,791.80	97.7%
SD (+/-)		n.a.		0.07		2.40		62.08	
CV (%)		n.a.		0.1%		0.1%		0.7%	
N		0		2		2		2	

Table 7 Accuracy and Precision in dog liver tissue

2439exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	Sample No.	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal
AN401.0-24_01_AB08	29, 41, 53, 65	19.1	95.5%	61.4	102.3%	4,500.0	100.0%	8,934.0	99.3%
	30, 42, 54, 66	19.9	99.5%	61.0	101.7%	4,503.8	100.1%	8,932.3	99.2%
	31, 43, 55, 67	21.4	107.0%	59.5	99.2%	4,495.2	99.9%	8,908.7	99.0%
Mean		20.13	100.7%	60.63	101.1%	4,499.67	100.0%	8,925.00	99.2%
SD (+/-)		1.17		1.00		4.31		14.14	
CV (%)		5.8%		1.7%		0.1%		0.2%	
N		3		3		3		3	

Table 8 Accuracy and Precision in dog kidney tissue

2439exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	Sample No.	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calc. 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal
AN401.0-24_01_AB08	33, 45, 57, 69	19.7	98.5%	60.3	100.5%	4,525.6	100.6%	8,923.8	99.2%
	34, 46, 58, 70	21.9	109.5%	61.0	101.7%	4,543.7	101.0%	8,955.3	99.5%
	35, 47, 59, 71	20.1	100.5%	60.7	101.2%	4,535.9	100.8%	8,901.9	98.9%
Mean		20.57	102.8%	60.67	101.1%	4,535.07	100.8%	8,927.00	99.2%
SD (+/-)		1.17		0.35		9.08		26.84	
CV (%)		5.7%		0.6%		0.2%		0.3%	
N		3		3		3		3	

Table 9 Freeze and thaw and short term stability in dog brain

2439exNA				QC-Low		QC-High	
				60.0 ng/g		9,000.0 ng/g	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439exNA Conc. ng/g**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439exNA Conc. ng/g**	**Acc. Criterion +/-15 % of Nominal**
AN401.0-24_01_AB09	21-Oct-2024	16, 20	4 hours at room temperature	59.0	98.3%	9,159.5	101.8%
	21-Oct-2024	17, 21		62.2	103.7%	9,164.3	101.8%
	21-Oct-2024	18, 22		63.6	106.0%	9,115.1	101.3%
Mean				61.60	102.7%	9,146.30	101.6%
SD (+/-)				2.36		27.13	
CV (%)				3.8%		0.3%	
N				3		3	

Table 10 Dilution Integrity in dog brain tissue

2439exNA		Dilution 1:1,000			
		5,000,000.0 ng/g			
		10 % ACN / 0.01% Tween 20 / 1x PBS / 1 mM TCEP		10 % ACN / 1x PBS / 1 mM TCEP	
Run ID	Sample No.	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal
AN401.0-24_01_AB08	82, 86	5,280,719.4	105.6%	4,853,162.7	97.1%
	83, 87	5,425,267.0	108.5%	4,832,023.6	96.6%
	84, 88	5,236,577.1	104.7%	4,731,685.4	94.6%
Mean		5,314,187.83	106.3%	4,805,623.90	96.1%
SD (+/-)		98,696.87		64,899.09	
CV (%)		1.9%		1.4%	
N		3		3	

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6 Records

The following records in addition to the above mentioned Forms, Master or Spreadsheet Applications are required in this process:

- N/A

7 Change History

Edition, Section(s)	Change	Rationale
Ed 01	New document	Assay description after the method development